

Alex Turianskyj

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🎓 EDUCATION

McGill University, B.Eng. in Software Engineering Co-op

GPA: 3.68/4.00 | Aug 2021 – Dec 2025

💻 SKILLS

Programming: TypeScript, C#, Python, Java, C++, SQL
Tools: Git, GitHub Actions, Docker, Azure, Vite, Bash, Jira
Testing: Cypress, Jest, JUnit, Pytest, Catch2, Selenium

Frontend: Angular (NgRx), React, Tailwind, HTML, CSS
Backend: Spring Boot, ASP.NET Core, PostgreSQL
Languages: English (native), French (native)

📁 WORK EXPERIENCE

Autodesk, Software Developer Intern | React, TypeScript, C++ Montreal, QC | May 2025 – Aug 2025

- Stabilized the **React** Properties panel for real-time collaborative editing in Fusion (200,000+ customers) by resolving 5+ UI regressions and increasing **TypeScript** test coverage using **Jest**.
- Augmented the **C++** deep-linking logic in the Fusion desktop client to support version history via URL parameters and implemented **Catch2** unit tests, enhancing cross-application navigation.
- Reduced incomplete panel analytics payloads by **80%+** by extending serialization for booleans and nested objects.
- Refactored native JSON event parsing with separate handling for model and component events, fixing a critical cache invalidation failure that affected property updates in configured designs.
- Integrated LaunchDarkly feature flags and **Cypress** E2E tests to enable a controlled rollout of a new panel layout.

Matrox, Software Engineering Intern | Angular, ASP.NET Core, C# Montreal, QC | Jan 2024 – Aug 2024

- Developed a **TypeScript** monitoring dashboard with live charting of device metrics to streamline remote analysis.
- Engineered a **C#** logging provider to transmit device logs via **SignalR**, accelerating root-cause troubleshooting.
- Optimized log architecture to increase retention by **75%** and support 3 independently filtered parallel log streams.
- Built **Angular** components from Figma designs with **NgRx** state management and extended the **ASP.NET Core** REST API for full-stack functionality, enabling key device features including volume and keyboard layout control.
- Established an automated end-to-end regression test suite using **JUnit** and **Selenium** to ensure build stability.

Hydro-Québec, Software Development Intern | JavaScript, VBA Montreal, QC | May 2023 – Aug 2023

- Automated the validation of substation data in Excel using **VBA** macros, reducing processing time by over **95%**.
- Engineered a **JavaScript** testing tool and designed a JSON schema for efficient storage and analysis of results.

🔗 SELECTED PROJECTS

Holoportation, C++, C#, Python, .NET, OpenCV [🔗](#) Sep 2024 – Apr 2025

- Built an AR system for HoloLens 2 astronaut training, named a **Top 5 Finalist** for the McGill Engine Capstone Prize.
- Enhanced LiveScan3D in **C++** and **C#** with WinForms for real-time 3D reconstruction using Femto Bolt cameras.
- Integrated **OpenCV** and YOLO-World to extract documents from depth-masked RGB streams for AR readability.

Daily Ball, Unity, C#, Android, WebGL [🔗](#) May 2023 – Jul 2024

- Developed a physics-based 2D mobile game in **Unity** and launched it on Google Play, achieving 1,000+ downloads.
- Increased player retention by **30%** through game updates and released a WebGL demo at **dailyball.alex.t.dev** [🔗](#)

Vibe, Python, Pandas, NumPy, SciPy, Streamlit [🔗](#) Sep 2023 – Nov 2023

- Created a content-based music recommender powered by a 1-million-song dataset, hosted at **vibe.alex.t.dev** [🔗](#)
- Compressed data by **75%** (400MB to <100MB) using **Pandas** and optimized recommendations to <5s with **SciPy**.

UniTrade, Java, Spring Boot, React, PostgreSQL, GitHub Actions [🔗](#) Jan 2023 – Apr 2023

- As Product Owner, led an 8-person **Agile** team to build a student marketplace using **Spring Boot** and **React**.
- Prioritized tasks, authored Gherkin user stories and implemented API endpoints with **JUnit** and **Mockito** tests.